

Total No. of Questions : 10]

SEAT No. :

[Total No. of Pages : 2

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[5871]-703

B.E. (Computer Engg.)

DATA ANALYSIS

(2015 Pattern) (Semester - I)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Use of Calculator is allowed.
- 5) Assume suitable data, if necessary.

Q1) a) Explain the major characteristics of Big Data. **[5]**

b) What is the hypothesis? Explain with a suitable example the term hypothesis testing? **[5]**

OR

Q2) a) Explain each phase in the data analytics life cycle? **[5]**

b) What is an analytic sandbox, and why is it important? **[5]**

Q3) a) Explain Student's t-test method with an example. In which condition will you select Student's t-test? **[5]**

b) What are the problems with the Apriori algorithm? How do we improve the efficiency of the Apriori algorithm? **[5]**

OR

Q4) a) What is ANOVA stands for? Explain the term ANOVA with a suitable example? **[5]**

b) Explain Support, Confidence, Lift, and Leverage, and also explain their importance in the association rule? **[5]**

P.T.O.

- Q5)** a) Explain the Decision tree algorithm with an example? [7]
b) What is data smooting? Explain the steps involved in the data smoothing process? [8]

OR

- Q6)** a) What is Gini Index? Explain in brief the CART algorithm? [7]
b) Write the mathematical equation for the Bayes theorem and explain it with a suitable example? [8]

- Q7)** a) What is meant by machine learning? Explain supervised - unsupervised learning algorithms. [7]
b) Explain in brief, different conventional data visualization tools? [8]

OR

- Q8)** a) What are the major applications of the association rule algorithm? Explain the reason for applying the association rule to the mentioned applications? [7]
b) Explain in brief, different conventional data visualization tools? What is data visualization? Explain the five main products of Tableau. [8]

- Q9)** a) Explain the major features of HBase? Discuss the applications where HBase was used? [5]
b) Explain Apache Hadoop & HDFS with a neat diagram. [7]
c) Explain how Apache Pig can be used to analyze large sets of data? [8]

OR

- Q10)** a) Differentiate between SQL & NoSQL. [5]
b) Explain the Storage Mechanism in HBase. [7]
c) Explain the component diagram which depicts the architecture of Hive? [8]
